Safe Havens in Equity Crashes: Gold, Long Treasuries and Bitcoin on the Worst US Equity Days

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the mean return of the asset on the worst-decile S&P 500 days is not positive; H_1: it is positive (safe haven), for gold, long Treasuries or Bitcoin

Data: daily returns of GLD, TLT and BTC-USD; crash threshold -1.10% (10th percentile of S&P 500 daily returns)

Method: One-sided recentred stationary bootstrap of the mean on crash days; Bonferroni $\alpha = 0.0167$ for three assets

Results: GLD mean -0.10% ($p = 0.8444$); TLT mean 0.25% ($p = 0.0108$); BTC-USD mean -1.72% ($p = 1.0000$).

Research decision: ACCEPTED

1 Design

A safe haven is an asset that is uncorrelated or negatively correlated with equities in market stress [Baur and McDermott, 2010]. Crash days are defined by the 10th percentile of the whole-sample S&P 500 return distribution, which is fixed before looking at the assets.

2 Results

Table 1: Mean asset return on the worst 10% of S&P 500 days

Asset	Crash days	Mean (crash)	Mean (all days)	One-sided p
GLD	252	-0.104%	0.050%	0.8444
TLT	252	0.254%	-0.015%	0.0108
BTC-USD	252	-1.718%	0.193%	1.0000

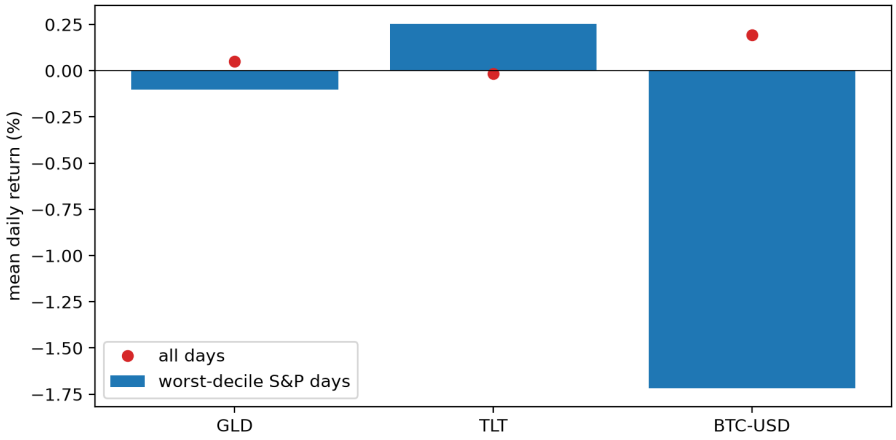


Figure 1: Mean daily return on crash days versus all days.

3 Limitations

Bitcoin trades around the clock and its daily close (00:00 UTC) is not synchronous with the US close, which can blur the same-day relation. The crash days cluster in a few episodes (2018, 2020, 2022), so the effective number of independent crashes is small.

References

Dirk G Baur and Thomas K McDermott. Is gold a safe haven? international evidence. *Journal of Banking and Finance*, 34(8):1886–1898, 2010.